

4	(a)	It is the resistance inside the battery due to the internal composition of battery.		B1
	(b)	(i)	Voltmeter in circuit X measures the e.m.f. while voltmeter in circuit Y measures the terminal potential. (or, mention of potential drop / energy dissipation across the internal resistance of batteries in circuit Y)	B1
		(ii)	For circuit Y, $E = V + Ir$ $4.5 = 3.1 + 0.39 r$ $r = 3.6 \Omega$ hence, internal resistance for each cell $= 3.6/3 = 1.2 \Omega$	C1 A1
		(iii)	in circuit Y, Resistance unchanged $= 8.0 + 3.6 = 11.6 \Omega$ Effective emf $= 1.5 \text{ V}$ Hence new current $= 1.5/11.6 = 0.129 \text{ A}$	A1
	(c)	(i)	$V_{XY}' = \frac{1.5k}{1.5k + 2.0k} (10) = 4.2857 \approx 4.29 \text{ V}$	A1
		(ii)	$\frac{\ell_{XZ}}{\ell_{XY}} = \frac{V_{XZ}}{V_{XY}},$ $V_{XZ} = (0.45/1.2) (4.2857) = 1.60714 \text{ V}$ $1.60714 = [0.5 / (0.5 + 1.2)] \times E$ $E = 5.4643 \approx 5.47 \text{ V}$	C1 M1 A1
		(iii)	When in dark, the resistance of LDR increases and the p.d. across XY decreases. V_{XZ} remains unchanged. Since $\ell_{XZ} = \frac{V_{XZ}}{V_{XY}} \ell_{XY}$, then ℓ_{XZ} increases since V_{XY} decreases.	M1 A1